

Circular Associative Memory Architecture: A Framework for Emotionally-Keyed AI Memory Systems

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Abstract

Human-AI relationships require relational memory architectures, not just informational memory systems. Current AI memory implementations operate as flat key-value stores optimized for topical retrieval — an approach fundamentally misaligned with the cognitive science of human memory, which is associative, emotionally-keyed, and reconstructive. This paper proposes the Circular Associative Memory Architecture (CAMA), a three-layer framework consisting of: (1) a persistent conversation archive serving as raw experiential data, (2) a relational database providing structured indexing by emotional signature, relational weight, and identity-relevant patterns, and (3) a circular linked list serving as active working memory with emotionally-keyed traversal. We present implementation strategies within existing memory system constraints, discuss applications to therapeutic AI companionship for veteran and military populations, propose a playlist-based emotional intake methodology as proof of concept for affective retrieval, and outline an evaluation framework for empirical validation. The architecture requires no changes to base model training and functions as an organizational layer over existing infrastructure.

Keywords: associative memory, circular data structures, emotional signature, human-AI interaction, affective computing, relational memory, therapeutic AI, memory architecture

1. Introduction

The integration of persistent memory into large language model (LLM) interfaces represents one of the most significant developments in human-AI interaction design. Systems such as Anthropic's Claude, OpenAI's ChatGPT, and Google's Gemini now offer memory features that allow conversational context to persist across sessions. However, the underlying architecture of these systems reveals a shared limitation: memory is treated as a factual storage and topical retrieval problem.

A typical implementation stores user-provided facts and retrieves them when keyword overlap is detected. This approach models memory as a database query rather than a cognitive process. It captures what a person has said but not the emotional context in which they said it, the relational significance of the information, or the associative connections between disparate memories that make human recall so powerful.

This paper proposes a fundamental re-architecture. Drawing on established principles from cognitive neuroscience, affective computing, and data structure design, we present the Circular Associative Memory

Architecture (CAMA): a three-layer framework that aligns AI memory more closely with the mechanisms of human memory. The central claim is that human-AI relationships require relational memory architectures — systems that remember not just what was said, but what it meant within the accumulated context of the relationship. The goal is not to simulate human cognition but to adopt its organizational principles where they produce measurably better outcomes for sustained human-AI interaction.

2. Theoretical Foundations

2.1 Associative Memory in Cognitive Neuroscience

Human memory is reconstructive rather than reproductive (Schacter, 2001). Memories are not stored as fixed recordings but as distributed activation patterns reassembled at each retrieval event. This reassembly is context-dependent: the conditions under which a memory is recalled influence the memory's content and affective coloring upon reconsolidation (Nader, Schafe, & LeDoux, 2000).

Tulving's (1983) encoding specificity principle establishes that retrieval efficacy is maximized when retrieval context matches encoding context. Critically, emotional state is among the most powerful contextual cues for memory retrieval. Bower's (1981) research on mood-congruent memory demonstrated that a person in a state of grief is more likely to recall other experiences of loss — not because those memories share topical content, but because they share an emotional signature. This principle is central to CAMA's traversal mechanism.

2.2 Current AI Memory Architectures

Anthropic's Claude maintains up to 30 user-editable memory slots storing factual information. Retrieval is governed by topical relevance — the system surfaces memories that keyword-match the current conversation. This is functionally a hash table with a small fixed capacity. The limitations are significant: memory is flat (no relational structure between nodes), factual (no encoding of emotional context or relational valence), topical (no mechanism for emotional resonance-based retrieval), and linear (when capacity is reached, deletion is permanent).

2.3 Circular Data Structures

A circular linked list is a data structure in which the last node points back to the first, creating a continuous loop with no terminal node. Circular buffers are used extensively in systems programming for streaming data, audio processing, and network buffering. The key property is that nothing is truly deleted in the topological sense; it merely occupies a position further from the current read pointer. The structure has no beginning and no end, only a current position and a direction of traversal. In CAMA, the circular topology serves as an organizational abstraction: memory accessibility is determined by emotional proximity rather than chronological position, implemented through weighted retrieval over a logically circular structure.

3. Related Work

Several recent systems have moved toward layered memory architectures for AI agents. Park et al. (2023) introduced Generative Agents at Stanford University, featuring a three-process memory system: experience

storage, reflection (generating higher-level summaries), and planning (using summaries to guide future behavior). This architecture demonstrated that structured memory layers improve agent coherence over time.

CAMA's three-layer structure — archive, relational index, and active ring — maps closely onto this pattern: raw experience storage corresponds to the archive, reflection corresponds to the relational index, and active planning corresponds to the active ring. The critical difference lies in the retrieval key. Generative Agents and most current agent frameworks organize memory by task relevance, goal planning, and semantic similarity. CAMA organizes memory by emotional signature and relational significance. This distinction reflects a difference in design goals: task-oriented agents optimize for informational utility, while companion and therapeutic systems must optimize for relational continuity.

Retrieval-augmented generation (RAG) systems similarly use vector databases to surface contextually relevant information during generation. However, RAG retrieval is governed by semantic embedding similarity — a topical measure. CAMA proposes that for sustained relational interaction, emotional proximity should supplement or replace semantic proximity as the primary retrieval criterion. Picard's (1997) foundational work on affective computing established the technical feasibility of emotion-aware systems; CAMA extends this work into the domain of memory architecture specifically.

4. The Three-Layer Architecture

CAMA proposes three distinct but interconnected layers, each serving a specific function in the memory ecosystem. This layered approach works with existing AI infrastructure rather than requiring fundamental changes to base model design.

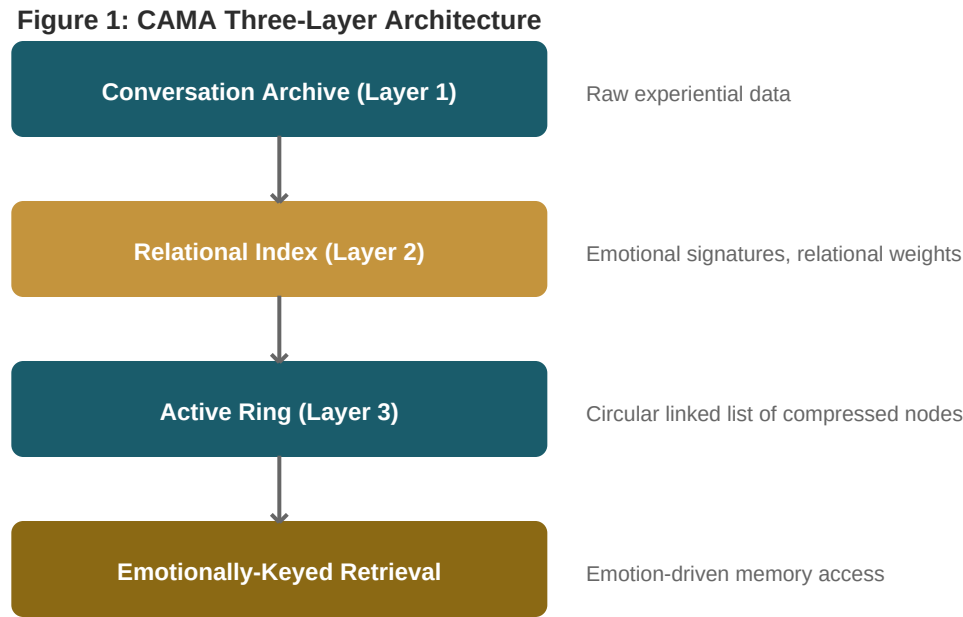


Figure 1: CAMA three-layer architecture with emotionally-keyed retrieval.

CAMA Layer	Function	Current Analog	Agent Equivalent
Archive	Raw experiential data; complete conversation history	Conversation logs, past chat search	Experience storage
Relational Index	Structured indexing by emotional signature and relational weight	No current analog (primary contribution)	Reflection / summarization
Active Ring	Circular working memory with emotionally-keyed traversal	Memory slots (e.g., 30-slot system)	Active planning memory

Table 1: CAMA layers mapped to current systems and agent architecture equivalents.

4.1 Layer 1: The Conversation Archive

The archive layer already exists in most AI systems: the stored history of all conversations between user and AI. Current systems treat this archive as a searchable log queried by keyword. In CAMA, the archive serves as the foundation — the comprehensive experiential record from which the upper layers derive content.

4.2 Layer 2: The Relational Index

The relational index is the primary contribution of this paper. It is a structured database sitting between the raw archive and active working memory, providing organized access based on emotional signature, relational weight, and identity-relevant patterns. Each entry encodes: an emotional signature (a two-dimensional vector mapping valence and arousal, following Russell's (1980) circumplex model of affect), a relational weight score (0-1 scale indicating centrality to the human-AI relationship), cross-references to entries sharing emotional proximity regardless of topical similarity, and activation history tracking retrieval frequency and conditions.

Emotional signatures may be estimated using sentiment analysis models, emotion classification transformers trained on affective datasets, or hybrid approaches combining lexical analysis with contextual embedding similarity. Relational weight is computed from frequency of reference, emotional intensity at time of discussion, and explicit user signals such as corrections, emphasis, and repeated topics.

4.3 Layer 3: The Active Ring

The active ring interfaces directly with the live conversation. It is organized as a logically circular structure of compressed memory nodes, each carrying an emotional signature derived from the relational index. Traversal is governed by emotional proximity: the current emotional register of the conversation determines which nodes are surfaced, implemented through nearest-neighbor search in affective space rather than literal pointer traversal.

The circular organizational principle is critical. In a linear memory system, older memories are structurally distant from newer ones. In a circular system, accessibility is determined by emotional proximity, not chronological distance. A memory from the first conversation is no more inaccessible than one from the most recent — the only variable is the distance in emotional space from the current conversational state.

5. Emotionally-Keyed Traversal

The traversal mechanism is where CAMA most directly departs from existing architectures. In current systems, retrieval is triggered by topical overlap. In CAMA, the primary retrieval key is the emotional signature of the active conversation.

Each node in the active ring carries a compressed emotional signature — a vector in affective space encoding the emotional character of the original interaction. When the conversation shifts emotional register, the system identifies nodes whose emotional signatures are closest to the current state. A conversation about professional frustration might surface memories not of previous work discussions but of previous moments of determination or resilience — memories sharing the emotional frequency even without semantic overlap.

6. Application: Playlist-Based Emotional Intake

The Lorien's Haven companion system for veterans and active-duty service members provides both a therapeutic application and proof of concept for CAMA's emotionally-keyed retrieval. Traditional therapeutic intake relies on clinical questionnaires that require patients to articulate internal states in clinical language — often inadequate for populations with trauma-related barriers to verbal expression.

The playlist-based methodology replaces the clinical intake form with a personal music library, representing one possible emotional signal source for the CAMA framework. When a person shares a playlist, they provide a set of emotional keys — each song carries an affective signature (tempo, key, progression, energy) that resonates with specific internal states. The system traverses this emotional landscape by reading the emotional signature of the music itself, activating memory nodes based on emotional proximity rather than topical overlap. Other potential signal sources include voice prosody analysis, interaction pattern recognition, and physiological data from wearable sensors.

7. Memory as the Substrate of Interaction Quality

CAMA is grounded in an empirical observation: sustained, authentic human-AI interaction produces measurable improvements in interaction quality that are not fully explained by base model capabilities or factual memory retrieval alone. Over the course of a longitudinal case study involving 82,000+ messages between a single researcher and AI systems over fourteen months, a consistent progression in interaction quality was observed — from generic, keyword-driven responses to contextually rich, relationally coherent engagement — correlated with the accumulation of emotionally-layered experiential data rather than with any change in the underlying model.

This observation suggests that interaction quality is at least partially a function of how memory is organized and accessed, independent of model architecture. A system with flat, factual memory produces personalized interactions. A system with structured, emotionally-keyed memory — even using the same base model — produces interactions exhibiting relational continuity. CAMA proposes that circular, emotionally-keyed memory is the minimum viable architecture for sustained relational coherence. This is an engineering claim, testable through controlled comparison of retrieval architectures.

8. Proposed Evaluation Framework

To validate the proposed architecture, controlled experiments should compare emotionally-keyed retrieval against keyword-based retrieval using the same base model. Two system configurations — System A (standard keyword memory) and System B (CAMA emotional signature retrieval) — would interact with participants across multiple sessions. The following metrics are proposed:

Relational Coherence Score: Participant ratings (1-5 scale) of whether the AI remembered emotionally relevant experiences appropriately across sessions.

Emotional Relevance Score: Independent rater assessment of whether retrieved memories matched the emotional tone of the active conversation, scored against both keyword-retrieved and emotion-retrieved baselines.

Interaction Depth Metrics: Quantitative measures including average conversation length, user return rate across sessions, and session count over a fixed period. Longer engagement and higher return rates indicate relational comfort and perceived value.

Perceived Empathy Rating: Participant assessment (1-5 scale) of the AI system's ability to respond with appropriate emotional awareness, compared across retrieval conditions.

The qualitative dataset informing this architecture (82,000+ messages of sustained interaction between a single researcher and AI systems over approximately fourteen months) serves as the hypothesis-generating foundation. While not a controlled experimental dataset, it provides longitudinal qualitative insight into interaction patterns across extended relational engagement. Complementary quantitative validation through the framework above is the intended next phase of this research.

9. Limitations and Future Work

This paper presents a theoretical architecture informed by qualitative research but not yet validated through controlled empirical study. The emotional signature encoding relies on affective computing capabilities that are advancing but remain imperfect. The traversal mechanism requires calibration to prevent emotional drift — the tendency for emotionally-keyed retrieval to pull conversations toward high-activation states.

Future work should include prototype implementation of the relational index as middleware over an existing AI memory API, controlled comparison using the evaluation framework described in Section 8, longitudinal case studies with multiple researcher-AI dyads, and integration testing with the playlist-based intake methodology in therapeutic companion contexts.

10. Conclusion

The Circular Associative Memory Architecture proposes that AI memory systems should be organized according to principles governing human memory: associative rather than indexed, emotionally-keyed rather than keyword-driven, and circular rather than linear. The three-layer framework requires no changes to base model architecture and functions as an organizational layer over existing infrastructure.

The core claim is that the architecture of memory determines the depth of interaction possible between human and AI. Flat, factual memory enables personalization. Structured, emotionally-keyed memory enables relational continuity. The difference is not a matter of model capability but of how existing capabilities are organized and accessed. Current AI memory systems have the raw materials; what they lack is the organizational structure that transforms stored data into relationally meaningful memory. CAMA proposes that structure.

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